TOR AGAINST THE QUOTIENTS BY POWERS OF AN IDEAL IS EVENTUALLY STRICTLY PRO-ZERO

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Artin-Rees for a submodule of a free module over a Noetherian ring. Algebra, Lemma \ref{algebra-lemma-Artin-Rees}, cited at proof lines 6467-6468. Bounds the intersection of the kernel with powers of the ideal inside a smaller power of the kernel.
- Dimension shift for Tor along a short exact sequence. used at proof lines 6470-6471. Reduces the higher degrees to the degree one case.

Source reference. Original lemma 0911: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11429–11429. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-quillenhomology: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6449–6449. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let A be a Noetherian ring. Let I be an ideal of A. Let M be a finite A-module. For every $p > 0$ there exists a $c > 0$ such that $Tor_p^A(M, A/I^n) \to Tor_p^A(M, A/I^{n-c})$ is zero for all $n \geq c$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Proof for $p = 1$. Choose a short exact sequence $0 \to K \to A^{\oplus t} \to M \to 0$. Then $\mathrm{Tor}_1^A(M, A/I^n) = K \cap (I^n)^{\oplus t}/I^n K$. By Artin-Rees (Algebra, Lemma 00IN) there is a constant $c \geq 0$ such that $K \cap (I^n)^{\oplus t} \subset I^{n-c}K$ for $n \geq c$. Thus the result for $p = 1$. For $p > 1$ we have $\mathrm{Tor}_p^A(M, A/I^n) = \mathrm{Tor}_{p-1}^A(K, A/I^n)$. Thus the lemma follows by induction. $\square$